Quality-Stratified Multimodal Fusion for Skin Lesion Malignancy Classification on MCR-SL

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Outline

Why This Problem?

- Deep learning has reached dermatologist-level accuracy on **image-only** skin lesion classification (Esteva et al., 2017; HAM10000).
- But a real dermatologist never diagnoses from an image alone — age, sex, lesion site, referral history all matter.
- A smaller line of work fuses images with **patient metadata** (e.g. MetaBlock on PAD-UFES-20) and shows it helps.
- **Missing piece:** none of these datasets record how *good* the diagnostic image itself was.
  - Real clinical images vary hugely in quality (lighting, focus, hair, blur).
  - No prior benchmark could ask: *do errors concentrate on poor-quality images?*

The Opportunity: MCR-SL

A brand-new dataset (Castro-Fernandez et al., *Data*, 2025) closes this gap:

- Every diagnostic image has an expert-assigned **1–10 quality rating**.
- Rich structured metadata per lesion/subject (22 fields).
- Partial histopathology confirmation (ground-truth beyond panel opinion).

Key fact

No prior published benchmark exists on MCR-SL. We are first.

This lets us ask a question no earlier skin-lesion benchmark could:
Does model performance degrade on lower-quality images, and can we train the model to be more robust to that?

MCR-SL Dataset at a Glance

Property	Value	Note
Subjects	60	59 have $\geq$1 binary-usable lesion
Lesions (total)	240	
malignant / non-malignant / unknown	42 / 192 / 6	18% malignant — heavily skewed
usable for binary task	**234**	240 − 6 `malignancy=="unknown"`
Images (raw rows in `image.xlsx`)	2394	
after lesion join + file check	**2131**	263 orphans dropped (21 lesion_ids)
dermoscopic (used for training)	**1352**	mean 5.73/lesion, range 1–18
clinical	779	not used — different visual domain
Expert quality ratings	1–10 scale	**the field no other dataset has**
lesions with a rating	238 / 240	3 experts (E001/E003/E004)
E002's ratings	entirely lost	0/241 — technical error, documented
Expert diagnostic certainty	0/25/50/75/100	all **4** experts intact (241/241)
Histopathology-confirmed	28 / 240	rest are expert-panel consensus
Metadata fields used (categ. / numeric)	16 / 4	missingness encoded, never imputed
Unified diagnosis classes	9	MEL=8, SCC=4, ANG=4, DF=2 (small-N)

"Usable" = has a lesion-record match and an existing image file on disk. Training is **image-level** (every dermoscopic image is a separate sample, ~5.7 per lesion); evaluation is **per-lesion** (one prediction each, from the `diagnosis_image_id` image).

Why the Analysis Tables Say $N = 231$, Not 234

A fair question: the quality terciles are $85 + 93 + 53 = \mathbf{231}$, but the binary task has **234** usable lesions.

Where did 3 go?

Count	**Step**	**Why**
240	All lesions in MCR-SL	
-4	no usable diagnosis image on disk	cannot produce a prediction at all
236	lesions with an out-of-fold prediction	
-5	`malignancy == "unknown"`	no valid binary label to score against
231	**prediction and binary label**	$= 85 + 93 + 53$ ✓

- **234** is a *labelling* count: $240 - 6$ lesions whose malignancy is "unknown".
- **231** is an *evaluable* count: lesions that have both a prediction and a label.
- The difference of **3** is binary-usable lesions whose diagnosis image is missing from disk — they are droppable at load time, logged, never silently filled in.

Two of the missing-image lesions (L0013, L0205) also have no diagnosis-image row at all, so they carry no quality rating either. **No lesion is ever excluded on the basis of its prediction** — only on missing input data or a missing label.

Task Definition

Primary task: Binary classification

Malignant vs. non-malignant, per lesion.
Ground truth: `unified_diagnosis` field — histopathology-confirmed where available, expert-panel consensus otherwise.

Secondary / exploratory task: 9-class diagnosis

NEV, SK, BCC, AK, ATY, MEL, SCC, ANG, DF.

Caveat: several classes have <10 lesions (MEL$=$8, SCC$=$4, ANG$=$4, DF$=$2) — reported as exploratory only, never as robust per-class statistics.

What This Work Does

1. First published benchmark on MCR-SL: image-only vs. late-fusion vs. **channel-gated fusion**, under a subject-disjoint 5-fold protocol.
2. **A quality-adaptive loss reweighting scheme** (`hard_mining`) that uses MCR-SL's expert quality ratings to weight the training loss — and a shuffled-quality control that validates the gain is genuinely quality-driven, not generic reweighting.
3. **Six robustness analyses** enabled by MCR-SL's quality ratings, expert certainty, histopathology subset, and multi-lesion subjects.
4. **Three apparent positive findings that we overturned ourselves** using proper controls — a validation discipline we report as part of the contribution, not an appendix.
5. Final result: **0.840 $\pm$ 0.095 balanced accuracy** (`hard_mining` + TTA + multi-image averaging), *within noise of* the closest verified binary comparator (0.836), on a dataset with $7\times$ fewer lesions.

The architecture is unchanged from standard components. The contribution is the *loss weighting* and the *validation*, not a new network.

Why MCR-SL, Not a Bigger Dataset?

The honest answer: it is not about scale. It is about a field that only exists here.

Dataset	Size	Expert per-image quality rating?
ISIC (2019/2020)	25,000–35,000+	No
HAM10000	10,015	No
PAD-UFES-20	2,298	No
MCR-SL	**240 lesions**	**Yes — 1–10, per diagnostic image**

- Larger datasets buy **statistical power**. None of them can answer this project's research question, because **the field the mechanism depends on does not exist in them**.
- MCR-SL's ratings are **real expert judgements of the photograph**, tied to the specific image used for each diagnosis — not an automated blur score or a learned proxy.
- We accept the cost of small N (wide error bars, reported throughout) in exchange for asking a question that is **unanswerable elsewhere**.

Has Our Research Question Been Answered?

The project asks **two separable questions**. They get two different answers.

Q1: Does performance degrade on lower-quality images? — No (null result)

Spearman(rating, error) $= -0.093$, $p = 0.158$, $N = 231$; tercile pattern non-monotonic (0.776 / 0.914 / 0.868). Reported as a **genuine null —** neither confirmed nor ruled out. **Not hidden.**

Q2: Can the model be trained to be more robust to quality? — **Yes, with evidence**

`hard_mining` loss reweighting: **+0.037** balanced accuracy, **+0.092** sensitivity, **+0.025** AUROC.

Validated, not asserted: a shuffled-quality control (same weight distribution, randomised lesion→rating mapping) **collapses the entire gain**.

Q1 being null does *not* invalidate Q2 — a model can be made more robust to a factor that shows no significant baseline correlation at this N.

Architecture — Two Encoders

Image Encoder

- EfficientNet-B0, ImageNet-pretrained
- Input: dermoscopic image, 224×224
- Output: 1280-d pooled vector *and* the last conv feature map ($1280 \times 7 \times 7$)

Metadata Encoder

- 16 categorical fields $\rightarrow$ learned embeddings (with explicit "unknown" category — *no imputation*)
- 4 numeric fields (age, height, weight, diameter) $\rightarrow$ z-scored using *train-fold-only* statistics + missingness bit
- Concat $\rightarrow$ 2-layer MLP $\rightarrow$ 128-d metadata vector

Both encoders feed into one of two fusion strategies (next slide).

Architecture — Two Fusion Strategies

1. Late fusion (baseline)

concat(pooled image vector, metadata vector) $\rightarrow$ Linear+ReLU+Dropout (256-d) $\rightarrow$ classifier.
This is the standard MetaBlock-style approach from prior literature.

2. Channel-gated fusion (our main method)

metadata vector $\rightarrow$ Linear $\rightarrow$ Sigmoid $\rightarrow$ a 1280-d **gate** $\in [0,1]^{1280}$
$\rightarrow$ multiply elementwise against the image's conv feature map, channel-by-channel
$\rightarrow$ global pool $\rightarrow$ same 256-d projection $\rightarrow$ classifier.

Idea: let patient metadata decide **which visual channels matter**, instead of just concatenating two vectors. Same mechanism as Squeeze-and-Excitation, but the gating signal comes from metadata, not from the image itself.

Architecture — Three Prediction Heads

All three heads sit on top of the **same** 256-d fused vector; only their outputs and losses differ.

1. Binary head (main task, always on)

256-d $\rightarrow$ Linear $\rightarrow$ 1 logit $\rightarrow$ sigmoid $\rightarrow$ P(malignant).
Loss: weighted BCE (or focal loss), class weights recomputed per training fold ($\sim$1:4.6 malignant:non-malignant).
This is the only head whose output is reported as the paper's headline results.

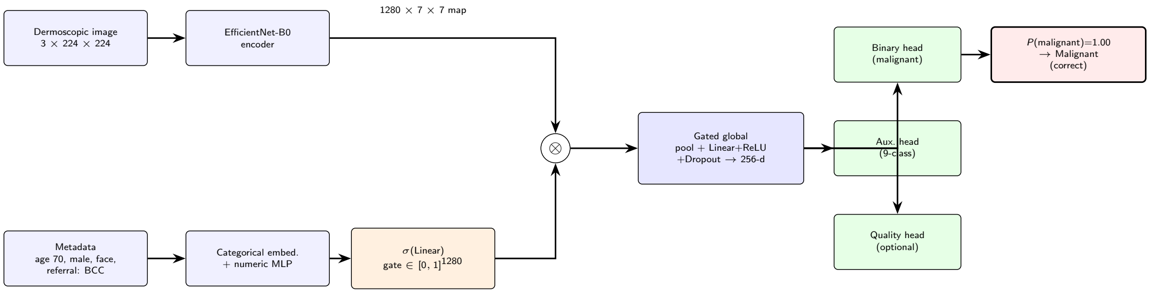
2. Auxiliary 9-class head (exploratory)

256-d $\rightarrow$ Linear $\rightarrow$ 9 logits $\rightarrow$ softmax $\rightarrow$ unified diagnosis (NEV, SK, BCC, ...).
Loss: class-weighted CE, downweighted to $\sim$0.3–0.5$\times$ the binary loss.
Reported only as an exploratory table — several classes have $<$10 lesions.

3. Quality-regression head (optional, quality-aware variant only)

256-d $\rightarrow$ Linear $\rightarrow$ 1 scalar $\rightarrow$ predicted mean expert quality rating (1–10). Loss: MSE, weighted 0.1–0.2$\times$.
Tests whether ***predicting*** *quality improves robustness — it did not. Later superseded by* ***weighting the loss*** *with quality instead, which did work.*

Full Pipeline (with a Real Example)



Example: lesion L0189, histopathology-confirmed basal cell carcinoma — one of the model's most confident correct predictions.

Training / Evaluation Protocol

Subject-disjoint stratified 5-fold cross-validation

- Split by `subject_id`, never by lesion — a subject can have multiple lesions, and images of the same lesion are near-duplicates.
- Folds balanced by greedily equalizing malignant-lesion counts per fold.
- For each rotation: 1 fold held out for **test**, 1 disjoint fold held out only for **checkpoint selection** (val), rest for training.
- **No hyperparameter tuning against final test-fold results.**

Losses

- Binary head: weighted BCE, class weights recomputed per training fold ($\sim$1:4.6 malignant:non-malignant overall).
- Auxiliary 9-class head: class-weighted CE, downweighted ($\sim$0.3–0.5$\times$), exploratory only.
- (Optional) Quality-regression head: low weight (0.1–0.2), predicts mean expert rating.

The 5 Folds, Visualized

60 subjects $\rightarrow$ 5 disjoint folds (balanced by malignant-lesion count). Every rotation uses a **different** test fold and a different, disjoint val fold — so all 5 folds eventually serve as the test fold exactly once.

Rotation	**Fold 0**	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**
1	Test	Val	Train	Train	Train
2	Train	Test	Val	Train	Train
3	Train	Train	Test	Val	Train
4	Train	Train	Train	Test	Val
5	Val	Train	Train	Train	Test

- **Train** folds: model weights are updated here.
- **Val** fold: used only to pick the best training checkpoint (never influences weights).
- **Test** fold: touched exactly once, only for final reported metrics — never used for any decision.

Final numbers = mean $\pm$ std of the 5 **test**-fold results (one per rotation).

Evaluation Metrics — What Each One Means

Metric	**Definition**	**Why it matters here**
Accuracy	(correct predictions) / (all predictions)	Can be misleading: predicting "non-malignant" always still scores $\sim$82%
Balanced accuracy	average of sensitivity and specificity	**Primary metric** — corrects for the malignant class being a minority ($\sim$18%)
Macro-F1	F1 (precision/recall harmonic mean) averaged equally across both classes	Rewards doing well on *both* classes, not just the majority one
Sensitivity (recall)	(malignant caught) / (all actual malignant)	Missed malignancy = missed cancer diagnosis — the costliest error clinically
Specificity	(non-malignant cleared) / (all actual non-malignant)	Low specificity means unnecessary biopsies/patient anxiety
AUROC	probability the model ranks a random malignant lesion above a random non-malignant one	Threshold-independent measure of overall ranking quality

All reported as mean $\pm$ std across the 5 test folds, plus a confusion matrix aggregated across all folds.

Core Ablation: Image-Only vs. Late Fusion vs. Channel-Gated

Metric	**Image-only**	**Late fusion**	**Channel-gated**
Accuracy	0.813 $\pm$ 0.020	0.831 $\pm$ 0.052	0.833 $\pm$ 0.070
Balanced acc.	0.784 $\pm$ 0.073	0.768 $\pm$ 0.090	0.783 $\pm$ 0.085
Macro F1	0.733 $\pm$ 0.038	0.743 $\pm$ 0.051	**0.757** $\pm$ **0.073**
Sensitivity	**0.697** $\pm$ **0.187**	0.644 $\pm$ 0.191	0.672 $\pm$ 0.159
Specificity	0.872 $\pm$ 0.061	0.891 $\pm$ 0.021	0.895 $\pm$ 0.048
AUROC	**0.895** $\pm$ **0.053**	0.851 $\pm$ 0.080	0.881 $\pm$ 0.065

Channel-gated wins macro-F1, but does **not** dominate every metric.

Image-only actually has higher sensitivity and AUROC — reported honestly, not hidden.

Balanced accuracy is effectively tied between image-only and channel-gated (0.784 vs. 0.783).

Quality-Aware Variant — A Negative Result

Added an auxiliary head to predict image quality, hoping it would make the model more robust.

Metric	Channel-gated	+ Quality-aware
Balanced accuracy	0.783	0.740
Sensitivity	0.672	0.578
Specificity	0.895	0.902
AUROC	0.881	0.879

It **underperformed** on 5 of 6 metrics — became more conservative (small specificity gain, big sensitivity loss). Worse direction for a cancer-screening task. Reported honestly as a negative result.

Follow-Up Intervention Sweep (11 configurations)

Configuration	**Acc.**	**Bal. acc.**	**Sens.**	**Spec.**	**AUROC**
Focal loss	0.858	0.809	0.694	0.923	0.877
Dermoscopy preprocessing	0.828	0.793	**0.736**	0.851	0.875
Contrastive auxiliary loss	0.812	0.770	0.669	0.872	0.873
Alt. optimizer (AdamW+cosine)	0.822	0.757	0.661	0.853	0.845
Test-time augmentation (TTA)	0.841	0.788	0.672	0.904	0.897
Multi-image averaging	0.854	0.813	0.719	0.907	0.906
Sharpness-aware min. (SAM)	0.858	0.810	0.694	0.927	0.904
SAM + multi-image	0.851	0.800	0.672	0.928	**0.917**
SAM + TTA	**0.861**	**0.822**	0.719	0.925	0.908
SAM + TTA + multi-image	0.856	0.810	0.694	0.925	0.915
Preprocess.+focal+SAM+TTA	0.818	0.759	0.622	0.895	0.893

Best in *this* sweep: SAM + TTA $\rightarrow$ 0.822. Stacking everything together **hurt** rather than helped. All of these were later superseded by the quality-adaptive loss (next section).

Quality-Adaptive Loss: The Mechanism

Every lesion has a mean expert quality rating $q \in [1, 10]$. We use it to **weight the training loss per sample** — not to predict it as an output (that was the auxiliary head, which failed).

`trust` — down-weight poor images

$$w = 0.5 + \frac{q-1}{9} \quad \Rightarrow \quad [1, 10] \to [0.5,\ 1.5]$$

"Low-quality images are less reliable, so trust them less."

`hard_mining` — up-weight poor images

$$w = 1.5 - \frac{q-1}{9} \quad \Rightarrow \quad [1, 10] \to [1.5,\ 0.5]$$

"Force the model to work harder on the difficult images."

$$\mathcal{L}_i = \underbrace{w_i^{\text{class}}}_{\text{existing class weight}} \times \underbrace{w_i^{\text{quality}}}_{\text{new}} \times \text{BCE}(\hat{y}_i, y_i)$$

Lesions with no valid rating get a neutral weight of 1.0. Fixed range, two directions, **no grid search** — the range was not tuned. Architecture, optimiser, folds and checkpoint rule are **identical** to the baseline; only the loss weighting changes.

Validating the Loss: The Shuffled-Quality Control

The obvious objection

"Maybe *any* per-sample reweighting acts as a regulariser. Perhaps the quality ratings are irrelevant and you have just found noise that happens to help."

The control: within each training fold, **shuffle** the quality ratings across lesions. Identical weight *distribution*, but the lesion→rating correspondence is destroyed. Test folds always use **real, unshuffled** ratings. Everything else identical.

Mechanism	**Bal. acc.**	**Sens.**	**High−Low gap**
Plain baseline (no quality)	0.783	0.672	0.091
`hard_mining` (real ratings)	**0.820**	**0.764**	0.091
`hard_mining` (shuffled)	0.775	0.667	0.016
`trust` (real ratings)	0.788	0.725	0.082
`trust` (shuffled)	0.763	0.672	0.021

- `hard_mining` **passes:** shuffling collapses it to 0.775 / 0.667 — *back to baseline*. The gain **requires the real quality signal**.
- `trust` **fails:** its apparently-narrower tercile gap (0.082) is *beaten* by the shuffled control (0.021). Revealed as a reweighting artifact — reported as a negative result.

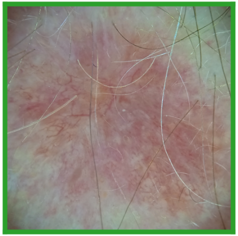
Where Does the Gain Actually Come From?

Decomposing the final result — **how much is genuinely about quality?**

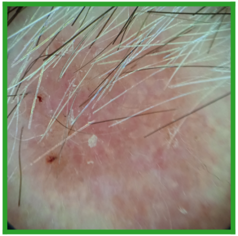
Step	**Configuration**	**Uses quality?**	**Bal. acc.**	**Δ vs base**	**Sens.**
0	Baseline (channel-gated)	No	0.783	—	0.672
1	+ `hard_mining` loss	**Yes**	0.820	**+0.037**	0.764
2	+ TTA (eval-time)	No	0.833	+0.050	**0.808**
3	+ multi-image (eval-time)	No	**0.840**	+0.057	0.783
ctrl	`hard_mining`, shuffled	Fake	0.775	−0.008	0.667

- Of the total **+0.057** gain, **+0.037 (≈65%)** is the validated, quality-specific contribution.
- The remaining **≈35%** is generic eval-time technique (TTA, multi-image averaging) with **nothing to do with quality** — stated explicitly so the novel part is not oversold.
- Steps 2–3 require **no retraining** at all.

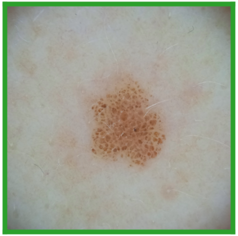
Example Predictions (Best Method)



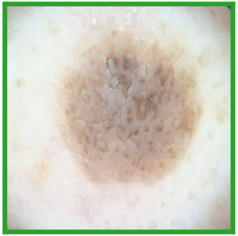
Age 70 / Male / Face
Referral: BCC
Ground truth: Malignant
Predicted: Malignant, P=1.00



Age 57 / Male / Face
Referral: BCC
Ground truth: Malignant
Predicted: Malignant, P=1.00



Age 50 / Female / Back
Referral: Voluntary sample
Ground truth: Non-malignant
Predicted: Non-malignant, P=0.00



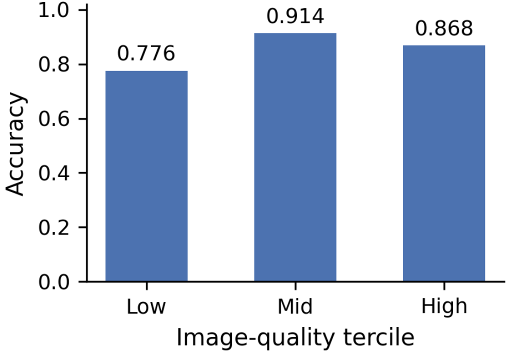
Age 64 / Male / Back
Referral: SK
Ground truth: Non-malignant
Predicted: Non-malignant, P=0.00

Highest-confidence *correct* predictions — illustrative only, not representative of overall performance (see full metrics above).

Why Robustness Analyses Are the Real Contribution

- Beating a prior number was never really possible — **no prior benchmark exists on MCR-SL**.
- What MCR-SL uniquely offers: per-image expert quality ratings + partial histopathology confirmation.
- So the paper's actual novelty is asking questions no earlier skin-lesion benchmark could ask:
  1. Does performance drop on lower-quality images?
  2. Can we train the model to be more robust to quality variation?
  3. Does performance differ on histopathology-confirmed vs. panel-consensus lesions?
  4. Does the model "look at" the same metadata fields that the dataset's own statistics flagged as significant?

1. Quality-Stratified Performance



Tercile	**N**	**Acc.**	**Sens.**
Low	85	0.776	0.667
Mid	93	0.914	0.733
High	53	0.868	0.500

Pattern is **not monotonic**. Spearman correlation (rating vs. error) $= -0.093$, $p = 0.158$ — not significant at $N = 231$. No strong quality-performance relationship detected (neither confirmed nor ruled out).

Robustness Across the Quality Ranking

Does the validated loss hold up **at every quality level**, not just on average?

Quality tercile	N	Accuracy plain	Accuracy hard_mining	Sensitivity plain	Sensitivity hard_mining
Low	85	0.776	0.776	0.667	**0.810**
Mid	93	0.914	0.903	0.733	0.733
High	53	0.868	0.868	0.500	**0.667**
High−Low accuracy gap		0.091	0.091		

- **The accuracy gap is unchanged** (0.091 → 0.091). `hard_mining` does *not* flatten the quality-tercile profile — stated plainly.
- **But sensitivity rises where it matters most:** on the *lowest*-quality tercile, malignant recall goes 0.667 → **0.810**. Under `hard_mining`, sensitivity is now **highest on the worst images** — exactly the on-mechanism behaviour up-weighting them should produce.

Honest framing: the *loss* improves quality robustness; the *architecture* is unchanged and we make no claim it is inherently quality-robust. Malignant counts within each tercile are small — directional evidence, not a decisive test.

2. Quality-Aware Training — Did It Help?

Tercile	N	Acc. (plain)	Acc. (quality-aware)
Low	85	0.776	0.729
Mid	93	0.914	0.903
High	53	0.868	0.830

- High−Low accuracy gap: **0.091 without** quality-awareness vs. **0.101 with** it.
- The intervention **widened**, rather than narrowed, the quality-performance gap.
- Direct negative result — reported as such.

3. Histopathology-Confirmed vs. Panel-Consensus

Group	N	Accuracy
Histopathology-confirmed	28	0.750
Panel-consensus-only	203	0.867

- Model performs *worse* on the histopathology-confirmed subset.
- Plausible reason: histopathology is ordered more often for clinically *ambiguous or suspicious* lesions in real practice — these are inherently harder cases.
- $N = 28$ is too small for a confidence interval — treated as a **qualitative** finding only.

4. Metadata Importance — Does the Model "Look at" the Right Fields?

Method: ablate each field (replace with "unknown"), measure change in output logit.

Highest ranked	**Importance**
Organ transplant history	0.046
Lesion status when captured	0.041
Referral diagnosis*	0.024
Sex*	0.015
Location group*	0.011
Lowest ranked	
Age / Weight / Diameter* / Height	$\leq$0.0011

* = field the dataset paper found significantly associated with malignancy.

Referral diagnosis, sex, location group align with the dataset paper. But **diameter** ranks lowest — likely an architectural artifact (12-d categorical embeddings vs. 1 scalar), not true unimportance.

Three Times We Almost Reported a False Positive

Each of these **looked like a result**. A control overturned it. We report all three.

What it looked like	**The check**	**What it actually was**
`trust` narrowed the quality-tercile gap 0.091 $\to$ 0.082 — a second quality-robustness win	Shuffled-quality control	Reversed. The *shuffled* control produced an even narrower gap (**0.021**). A generic reweighting artifact, not a quality effect.
Expert diagnostic certainty predicted model error, $\rho = -0.175$, $p = 0.008$ — apparently significant	Stratify by malignant / non-malignant class	Reversed. Dissolves within both strata ($p = 0.083$ / $p = 0.559$). Malignant lesions cluster at low certainty, so the pooled correlation was **class-mix confounding**.
"Train on all images per lesion, not just one" — a large untapped data opportunity	Verify the loader before building anything	False premise. Image-level training had been the default since run one (5.65$\times$ more samples than lesions). **Caught before any effort was wasted.**

At $N = 234$ with $\sim$8–9 malignant lesions per fold, **an uncontrolled first look reliably produces plausible positives**. We treat catching them as part of the contribution — and it is the most transferable lesson here.

What Is Our Novelty, Precisely?

Not the architecture. Not a new loss function class. Not state of the art.

1. **First use of real expert-assigned quality labels for quality-adaptive loss reweighting in medical image classification.** The closest precedents — **AdaFace** (CVPR 2022) and **MagFace** — come from face recognition and must use *feature-norm as a proxy*, because face datasets carry no expert quality annotation. MCR-SL does.
2. **The opposite weighting direction works here.** AdaFace/MagFace *de-emphasise* low-quality samples; our `hard_mining` *up-weights* them, and that is the variant that wins.
   Plausible mechanism: a dermatologist can often still diagnose correctly from a poor photograph, so the label remains valid and the hard case is worth learning from. In face ID, poor quality can make identity genuinely **unrecoverable** — so de-emphasis is right there, and wrong here.
3. **Distinguished from the nearest skin-lesion prior work.** `arXiv:2308.02119` (HAM10000) performs expert-informed sample reweighting, but weights by **how the label was confirmed** (diagnostic method). We weight by a **direct expert rating of the photograph itself** — a different signal answering a different question.
4. **The validation discipline is part of the claim.** The shuffled-quality control, and the three reversed findings, are reported as contributions — not buried.

Comparison with the Literature

Source	Task	Eval unit	Bal. acc.
This work (channel-gated baseline)	binary	234 lesions	0.783 ± 0.076
This work (+ `hard_mining`)	binary	234 lesions	0.820 ± 0.078
This work (+ TTA + multi-image)	binary	234 lesions	$\mathbf{0.840 \pm 0.095}$
DiffMIC, PAD-UFES-20 (2025)	binary	2298 images	0.836

This is NOT a claim of beating the state of the art

Different dataset, different institution, no shared test set. Our fold-to-fold std ($\pm$0.078–0.095) is **over 20$\times$ larger** than the 0.004 nominal gap — **statistically indistinguishable**. The last +0.020 came from *eval-time averaging*, not a better method.

Within noise of the closest verified binary comparator — see the next slide for why "*N*" is not directly comparable.

"234 vs. 2298" Is Not a Like-for-Like Comparison

The two numbers on the previous slide count **different things**. Ours is a *lesion* count; theirs is an *image* count.

Quantity	**MCR-SL (ours)**	**PAD-UFES-20**	**Ratio**
Images available	2,131	2,298	1.1×
Images used to train	**1,352** (dermoscopic)	**2,298**	**1.7**×
Lesions	**234** usable	∼**1,641**	**7.0**×
Subjects / patients	60	∼1,373*	∼23×
Evaluation unit	**per lesion**	**per image**	—
Malignant	42 lesions (18%)	*	—
Non-malignant	192 lesions (82%)	*	—

- We train on **1,352 images**, not 234 — every dermoscopic image of a training lesion is a separate sample (∼5.7 per lesion). "10× less data" overstated our disadvantage; removed.
- **The defensible statement is** 7× **fewer lesions**, with *comparable* image counts. Our 60 subjects vs. ∼1,373 patients is the largest real gap.
- Our 18% malignant rate is **heavily skewed** — exactly why we report *balanced* accuracy.

*
PAD-UFES-20 patient count and cancer/non-cancer split not independently re-verified — confirm against Pacheco et al. (2020) before submission.

Cross-Dataset Check: DeepDRiD (Not Completed)

What we set out to test

Does `hard_mining` transfer to a **second medical imaging domain** with a differently-structured quality annotation? **DeepDRiD** (diabetic retinopathy, ISBI 2020) ships per-image quality annotations alongside DR severity grades — the closest analogue to MCR-SL's ratings outside dermatology.

Status: gate failed — no results, and none estimated

A deliberate go/no-go gate ran **before** any training code was written, and it **failed**:

- The transfer was attempted and **aborted after 50 minutes** — `RPC failed (result=92)`, `early EOF`, `index-pack failed`. All 2,916 objects compressed server-side; the connection could not sustain a single stream long enough to deliver the pack.
- Measured throughput **40–64 KB/s** single-stream. Parallelism reaches ∼700 KB/s, but `git` is single-stream and the archive endpoint **ignores HTTP range requests**, so neither path can be parallelised. `git-lfs` is also unavailable.

No DeepDRiD number is reported, estimated, or implied anywhere in this deck.

Concrete future work. Having a gate that *can* fail — and honouring it — is the same discipline as the three reversed findings.

Limitations

- **Effect size vs. uncertainty:** our fold-to-fold std ($\pm$0.078–0.095 balanced accuracy) is **larger than most differences we report**. The `hard_mining` effect (+0.037) is supported by the shuffled control, but **no two configs within $\sim$0.05 should be called a win** over one another.
- **Single seed**, single institution, 234 usable lesions, $\sim$8–9 malignant lesions per test fold — one lesion flipping moves a fold's sensitivity by $\sim$0.11.
- **Folds are subject-imbalanced** (5 / 6 / 16 / 16 / 16 subjects) — balanced on malignant-lesion count, not on subject count.
- **No external validation.** The DiffMIC comparison is cross-dataset context, not a benchmark. The DeepDRiD cross-domain check **was not completed**.
- Ground truth mostly panel-consensus (28/240 histopathology-confirmed); several 9-class categories have $<$10 lesions — exploratory only.
- Quality terciles are modest ($N \approx$53–93); **six analyses were run with no multiple-comparison correction**.
- Metadata-importance ranking is confounded by embedding-dimension imbalance (12-d categorical vs. 1 scalar numeric).
- A known `pos_weight` miscalibration ($\sim$16%, lesion-level weight applied to image-level batches) is **quantified and disclosed**, not silently corrected.

Conclusion & Future Work

What we found

- **First benchmark on MCR-SL.** Final: $\mathbf{0.840 \pm 0.095}$ balanced accuracy — *within noise* of the closest verified binary comparator, on $7\times$ fewer lesions.
- **Q1 (does quality degrade performance?): null** — no significant relationship at $N = 231$.
- **Q2 (can we train for quality robustness?): yes** — `hard_mining` gives $+0.037$ balanced accuracy / $+0.092$ sensitivity, and **a shuffled control proves the gain depends on the real ratings**. Sensitivity on the *worst* images rises $0.667 \rightarrow 0.810$.
- **Three apparent positives we overturned ourselves**, plus quality-aware training, contrastive loss and alternative optimisers — all reported as negatives.

Future work

Complete the **DeepDRiD** cross-domain check; **multi-seed averaging** to shrink error bars that currently exceed most effect sizes; correct the disclosed `pos_weight` miscalibration; use the 779 unused clinical images with a modality indicator.

Thank you

Questions?